

Screen Sketches

Team 2 _mstafa_2

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Actors

1. USER

Upon logging in with a non-admin account, Users are given an initial page that will allow them to either perform edits on their profile, or to continue on to the game. Upon picking “Play Game”, the User will be inserted into a lobby, where they will be able to choose from 1 of 3 options:

a. Player

i. The Player’s capabilities can be split into the following categories:

1. Interaction

- a. The player will be able to control a sprite upon the screen.
- b. The player will have a score counter unique to them.
- c. The player will be able to satisfy a win condition.

2. Visuals

- a. The player can only see their sprite and the objects generated by DK.
- b. The player will be able to see every other player’s score in real-time.
- c. The player can see the required win condition and their progress towards it.

3. Data

- a. The player’s data will be saved to their unique account in the form of:
 - i. Number of WINS
 - ii. Highest SCORE
 - iii. ...

b. DK

i. DK’s capabilities can be split into the following categories:

1. Interaction

- a. DK can generate (or “throw”) objects that will traverse the screen as obstacles for Players. These obstacles, upon touching a player either decrement their score or eliminate them.

2. Visuals

- a. DK can see each player and their position with minimal (<1sec) lag
- b. DK can see each player’s score and how close the players are to winning

3. Data

- a. A USER playing as DK will also have their DK statistics uploaded to the server and tied to the USER’s account.

c. Spectator

2. ADMINISTRATOR

The ADMINISTRATOR account, upon logging in, will have all the permissions of a regular USER, in addition to the capabilities of performing CRUDL operations on all USERS in the database:

- a. They will be able to generate a list of currently logged in USERS, their roles, their statistics, as well as other information contained in the database.
- b. They will be able to manipulate this information.
- c. They will be able to stop games prematurely.
- d. They will be able to manipulate game rules.

Non Functional Requirements

1. Product Requirements
 - a. Usability
 - i. The application must be user friendly
 - ii. It must be able to run on any reasonable device
 - iii. It must be able to support 4 or more USERS at a time
 - iv. It must be able to play audio during gameplay
 - b. Efficiency
 - i. The application must be able to have minimal lag between client and server
 - ii. CRUDL Operations performed by ADMINs must also take minimal time
 - c. Reliability
 - i. The application must be able to maintain a consistent connection to the server database.
 - ii. The application must be able to maintain a consistent connection to the client.
 - iii. Players must feel that their movements correspond accurately to the inputs they put in.
2. Organizational Requirements
 - a. Delivery
 - i. The application must be delivered on time by DEMO 4
 - b. Implementation
 - i. The application must be built on Android Studio
 - c. Standards
 - i. The application must meet minimum requirements to be fully utilized and played consistently.

Tables and Fields

User Table: one to many

User ID: Int (Primary key)

Username: Char

Password: Char

Email: Char

User Type: ENUM (Player, Spectator, Admin)

User ID	Username	Password	Email	User Type
1	Null	Null	Spec1@domain.com	Spectator
2	Play1	Pass123	Play1@domain.com	Player
3	Admin1	AdminPass1	admin1@domain.com	Admin

Player File:

Num: Int (Foreign key)

Additional Info: Char (any other info on users we may need)

Obstacle Table: many to many

Obstacle ID: INT (Primary Key)

Obstacle Name: CHAR (ex, "Barrel", "Bomb")

Type ID: INT (Foreign Key) (referencing Obstacle Type (Type ID))

Position X: FLOAT (on screen)

Position Y: FLOAT (on screen)

Width: FLOAT (obstacle width)

Height: FLOAT (obstacle height)

Is Movable: BOOLEAN (False for static obstacles)

Speed: FLOAT (if Is Movable = True, then how fast)

Direction: CHAR (move "Right to Left")

Time Delay: INT (Min delay (in milliseconds) between obstacle appearances)

Last Spawn Time: DATETIME (Previously displayed on screen) (optional)

Obstacle Type Table:

Type ID: INT (Primary Key) (Unique ID for each type of obstacle)

TypeName: CHAR (ie, "Fixed", "Moving")

Subtype Name: CHAR (subtype "Barrel", "Bomb")

Action: CHAR (player should "Jump", "Run", "Duck")

Moving Obstacle Info Table:

Moving Obstacle ID: INT (Primary Key) (Unique ID for the moving obstacle)

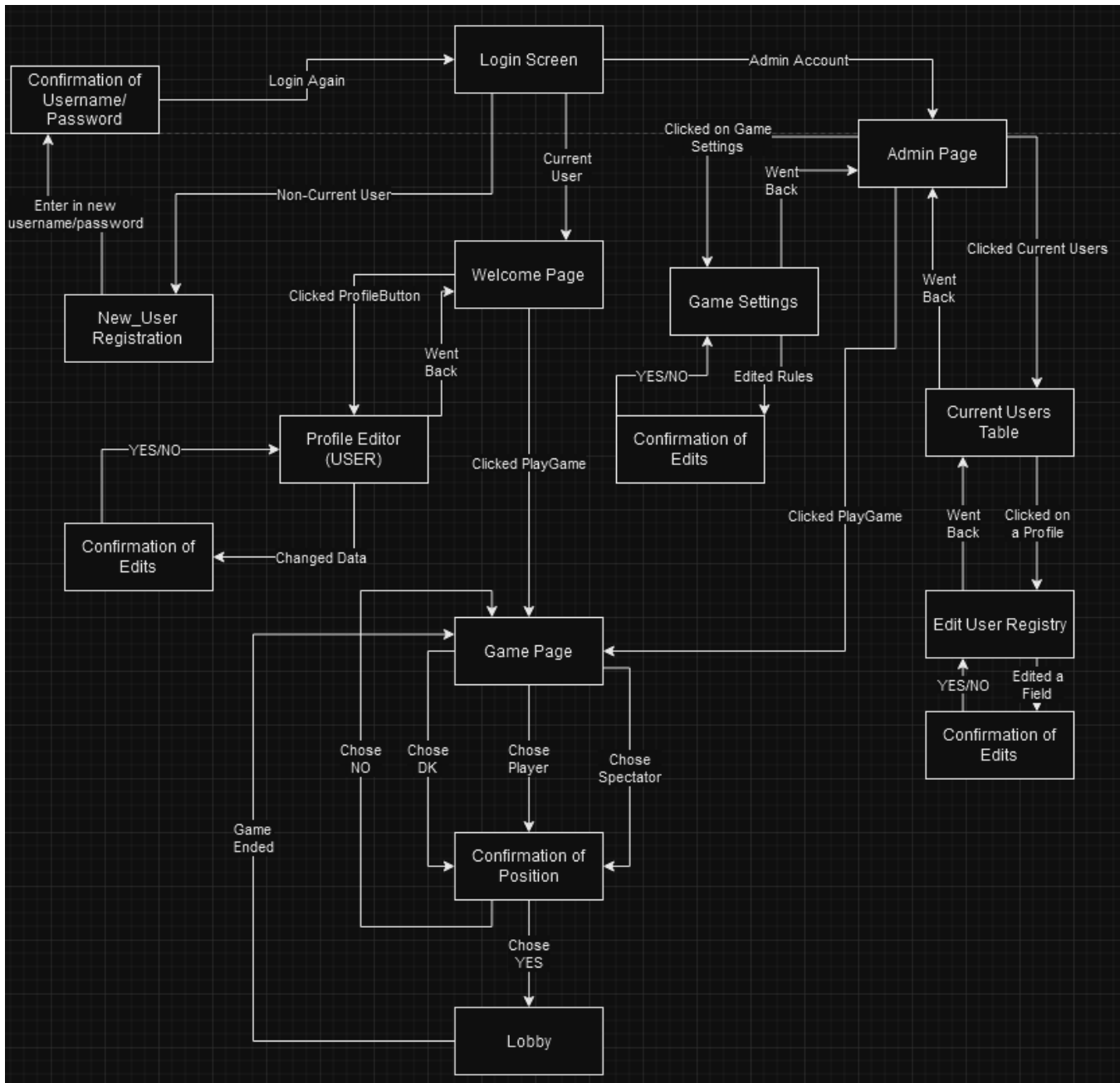
Obstacle ID: INT (Foreign Key) (referencing Obstacle (Obstacle ID))

Speed: FLOAT (how fast obstacle moves)

Direction: CHAR Direction of movement (move, "Right to Left", "Up Down")

Behavior Pattern: CHAR (ie., "Zig-Zag", "Straight")

Screen Flow Diagram



SCREEN SKETCHES

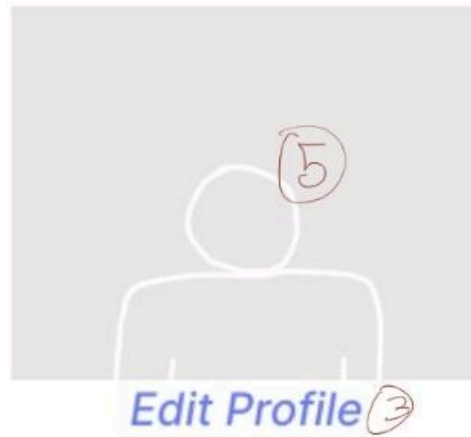
adminPage

Developed by: Sam Gumm

Welcome "USER_NAME"!

Admin_Operations:

- View Current Users ①
- Edit Game Rules ②



Actions:

- 1) The ADMIN is able to click upon the hyperlink, and a table will be created that shows all current users in the database along with their corresponding information.
- 2) The ADMIN is able to click upon the hyperlink, and a drop down menu will be displayed that allows tweaking of specific game rules, this is permanent until changed again.
- 3) The ADMIN can edit their profile by clicking this hyperlink, they will be taken to a separate page to perform edits.

4) The ADMIN can move to the Character Select page by clicking on the PLAY button. This is the page that will be displayed upon an ADMIN user logging in with the correct credentials. Normal USERS will be greeted with a similar page, but without administrative functionality.

lobbyPage

Developed by: Sam Gumm

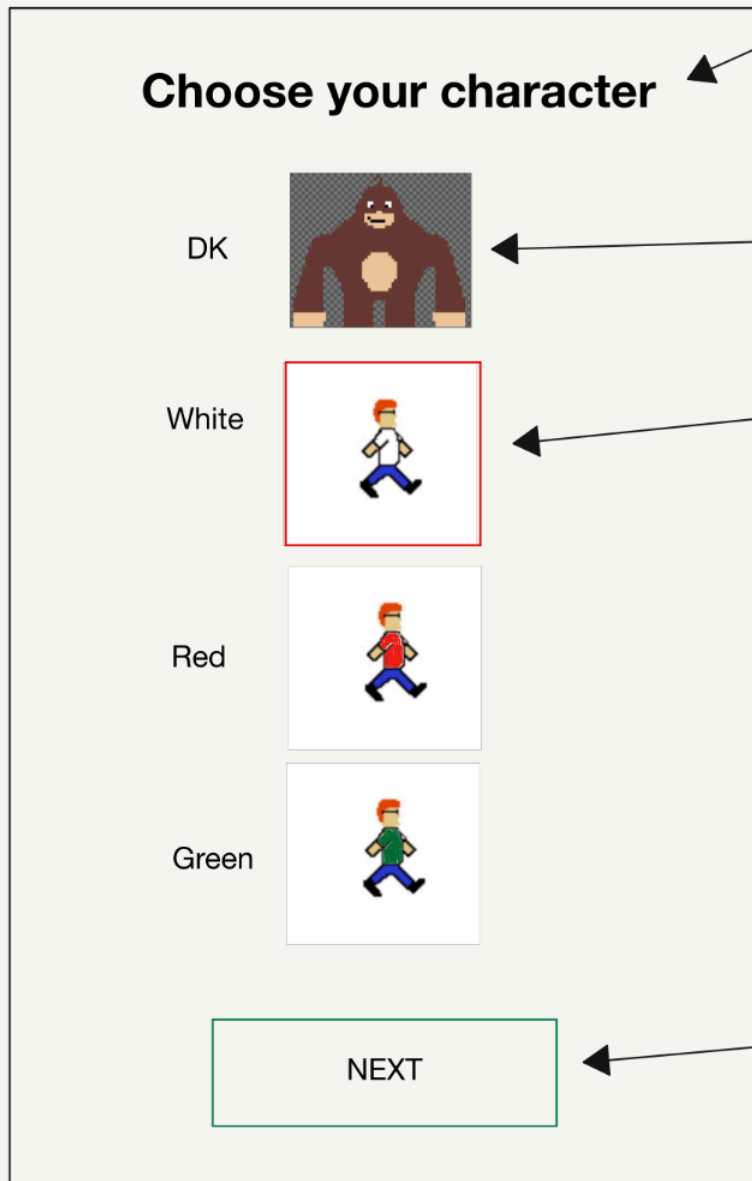


Actions:

- 1) The current game rules will be displayed here in the lobby. These are read-only, and there to inform players what to expect.
- 2) There are 4 slots in the lobby, which will be grayed out if there is not a player filling the slot. Each card displays information about the player.
- 3) There is a READY check mark button to indicate to other players who is ready and who is taking too long. Upon being clicked, a green checkmark is displayed to all other players in the lobby.
- 4) Each player's chosen character is indicated as well, if they are DK it will be indicated differently than if they are a RUNNER.
- 5) The RUNNER will be the same color as the sprite the player has chosen.
- 6) To play the game, all players must be ready for the PLAY button to become usable. Once it is clicked, all players will be taken to the Game Screen.

Character Selection Page

Developed by: Nicholas Doty

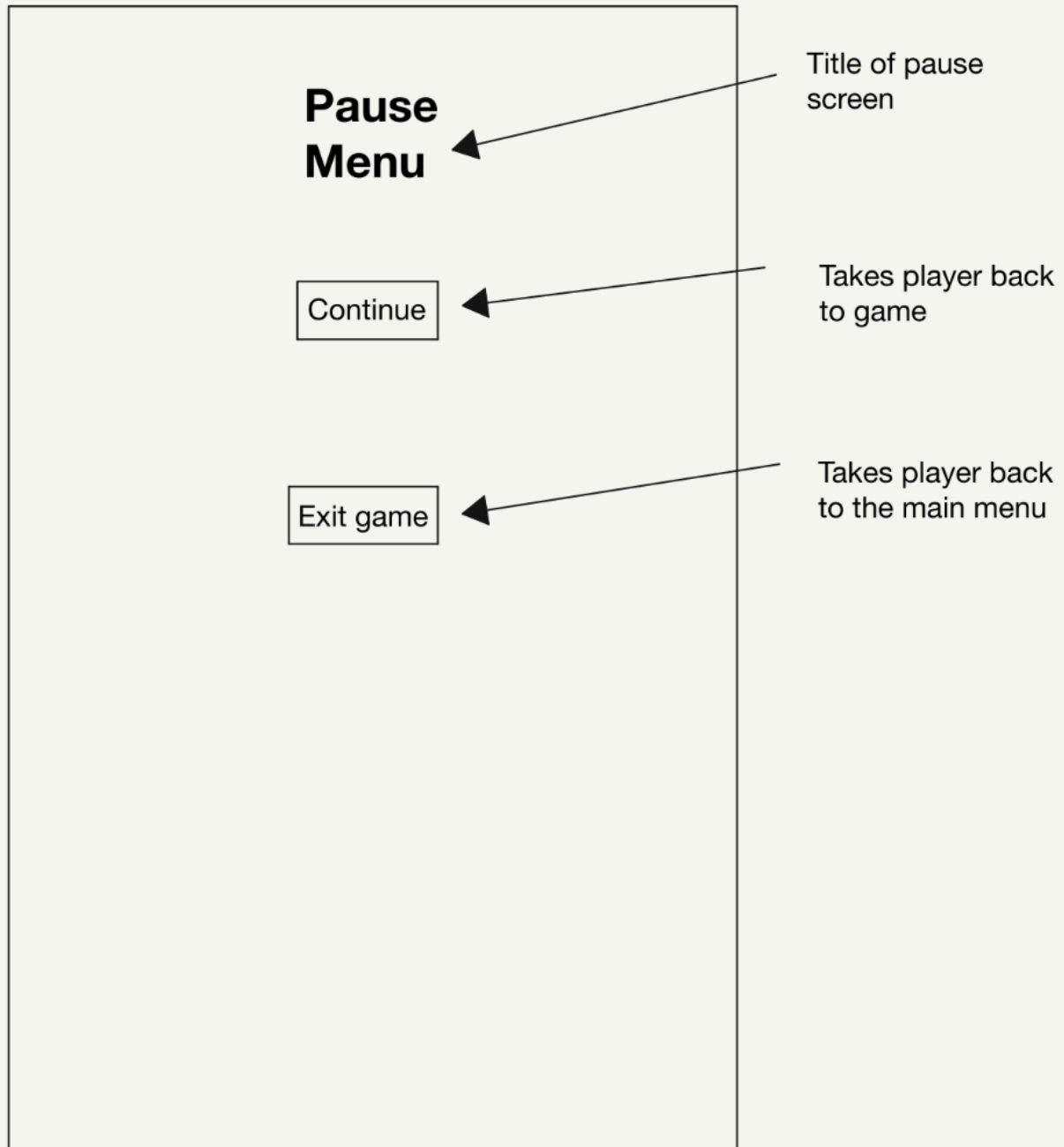


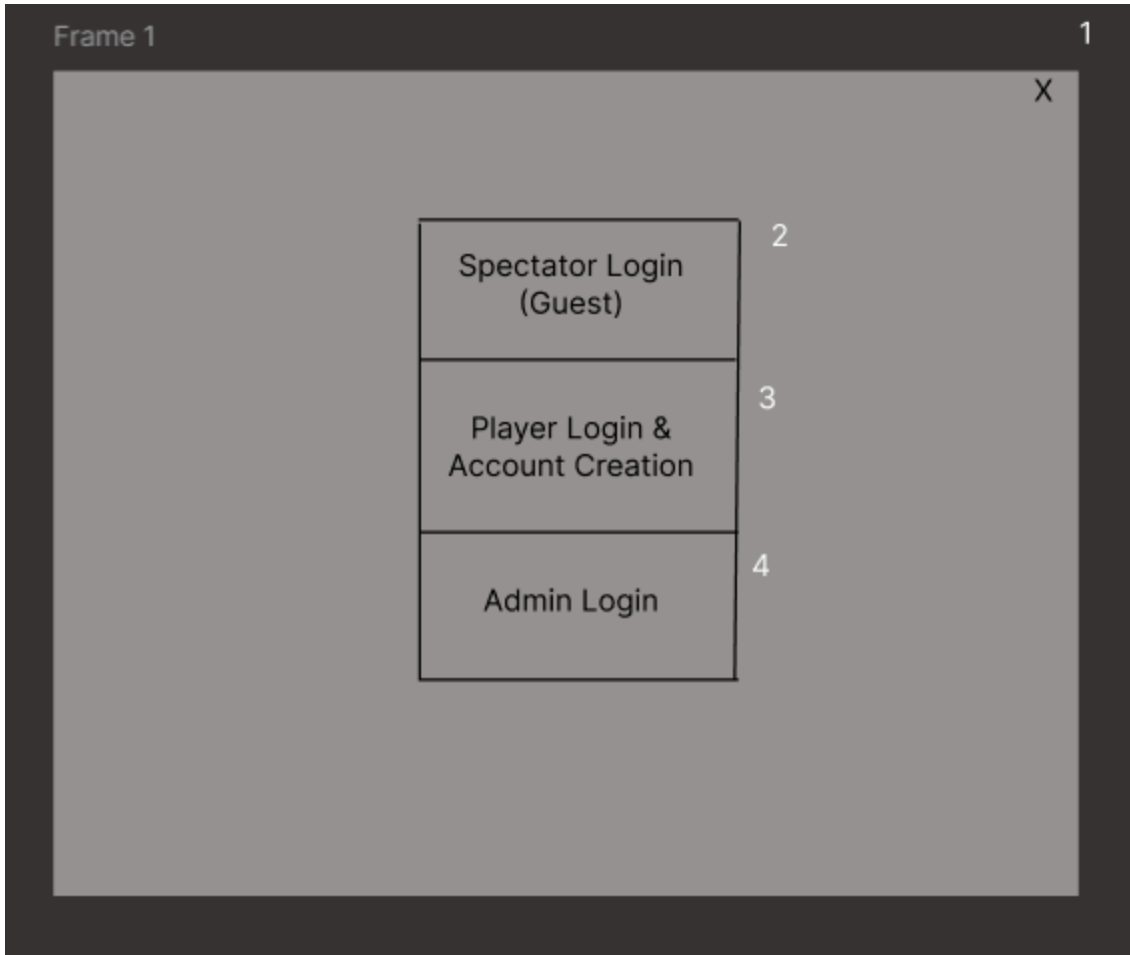
Choose between DK, white, red, or green.

DK can only be selected when there is more than one user playing. Otherwise he is removed or greyed out.

Once you click on the sprite of your choice, a box will highlight your selection. If another player has already chosen that sprite, it will be greyed out.

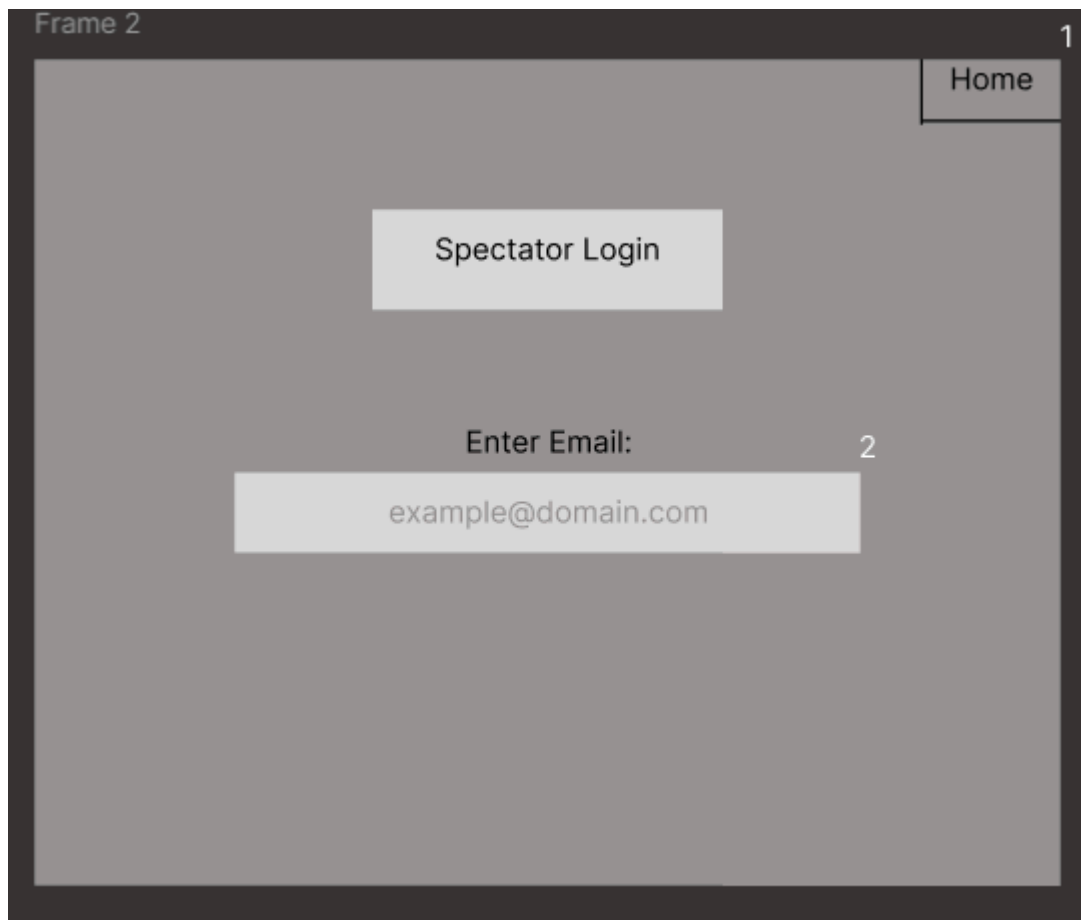
After choosing your selection, click next to take you to the confirmation of selection page.





- 1) This is the exit app button
- 2) Click here to begin the spectator type login process
- 3) Click here to begin the player login, will also be account creation page
- 4) Click here to begin the admin login process

This is where users will be taken to when they open the app (Home page). They can exit the app by clicking x or select the type of user they are and proceed with that login process and utilize the features specific to that login type. The exit app button only exists on this page for every other page there will be a return to home button



1) This is the return home button, should the user want to try a different login type

2) This is where the user enters their email to login as a spectator

Spectator login screen. The text box will clearly let the user know what login page they are in. As a spectator the user is not expected to create a username or password. Their email will not be saved as someone who has an account or be checked to see if an account exists with that email. Should they want to play the game they will need to use the player login and either enter their username, email, and password or create an account.

Frame 3

Home 1

Player Login

Enter Email: 2

example@domain.com

Enter Username: 3

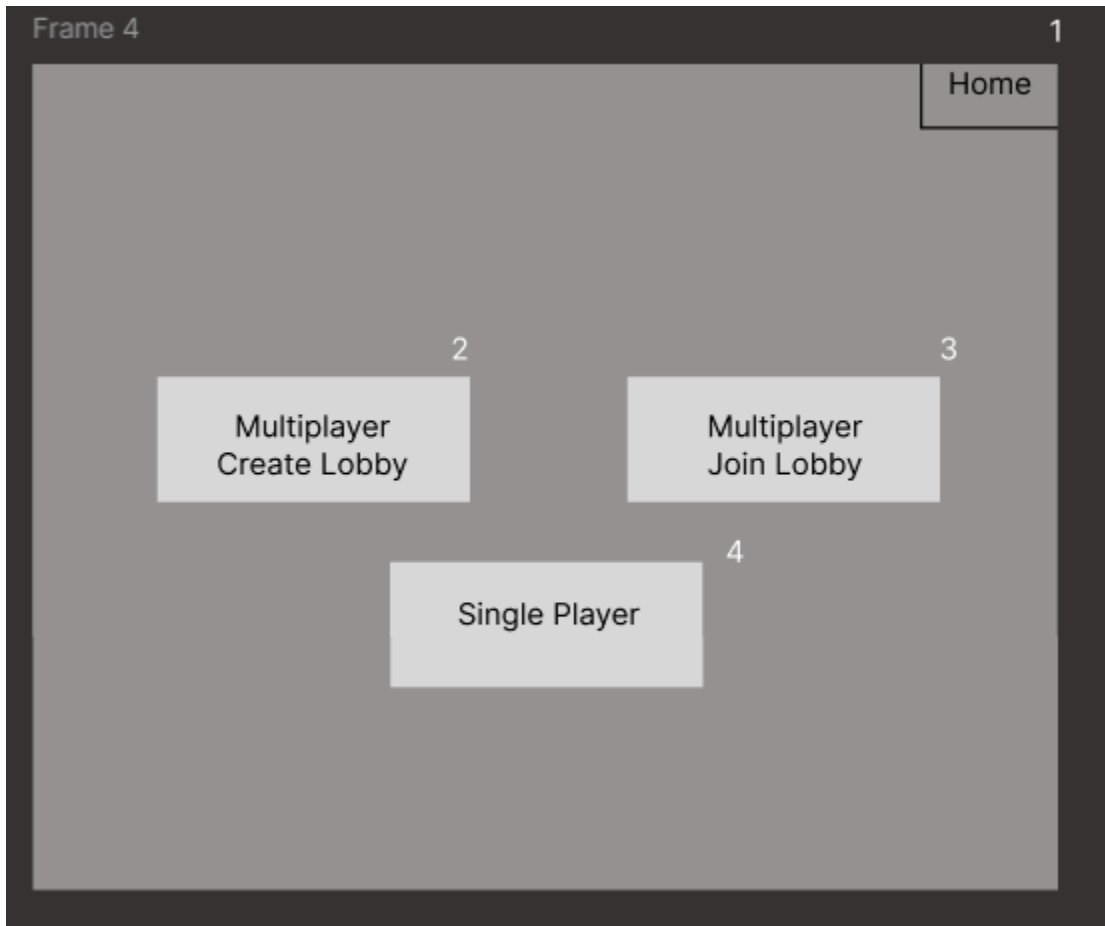
Ex: Player1

Enter Password: 4

Ex: PassWord123

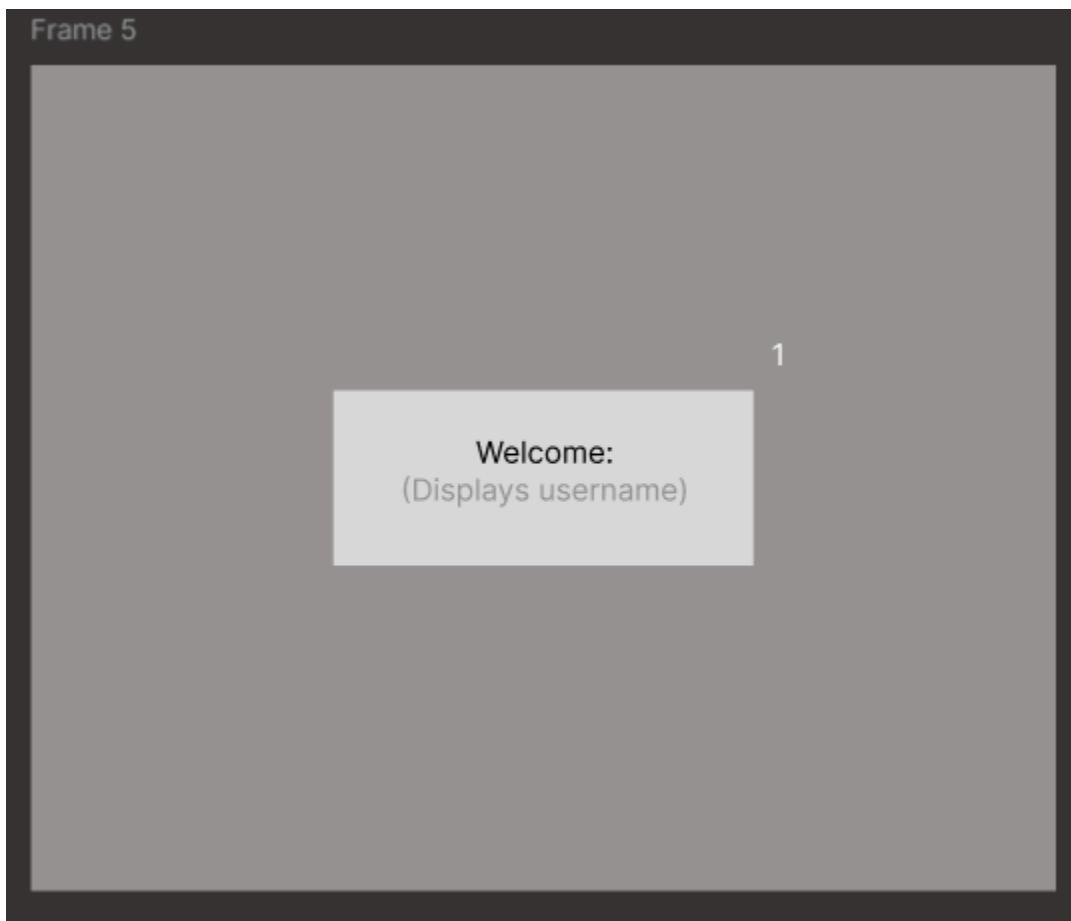
- 1) This is the return home button, should the user want to try a different login type
- 2) This is where the user will enter their email address
- 3) This is where the user will enter their username
- 4) This is where the user will enter their password

Player login page. The text box makes it clear what login page this is. The light color text shows what the format might look like. If all three do not match an existing account and no such email or password are already present then a new account is made, else there is a login error/response as this is not a new account and thus the info is probably entered wrong. .

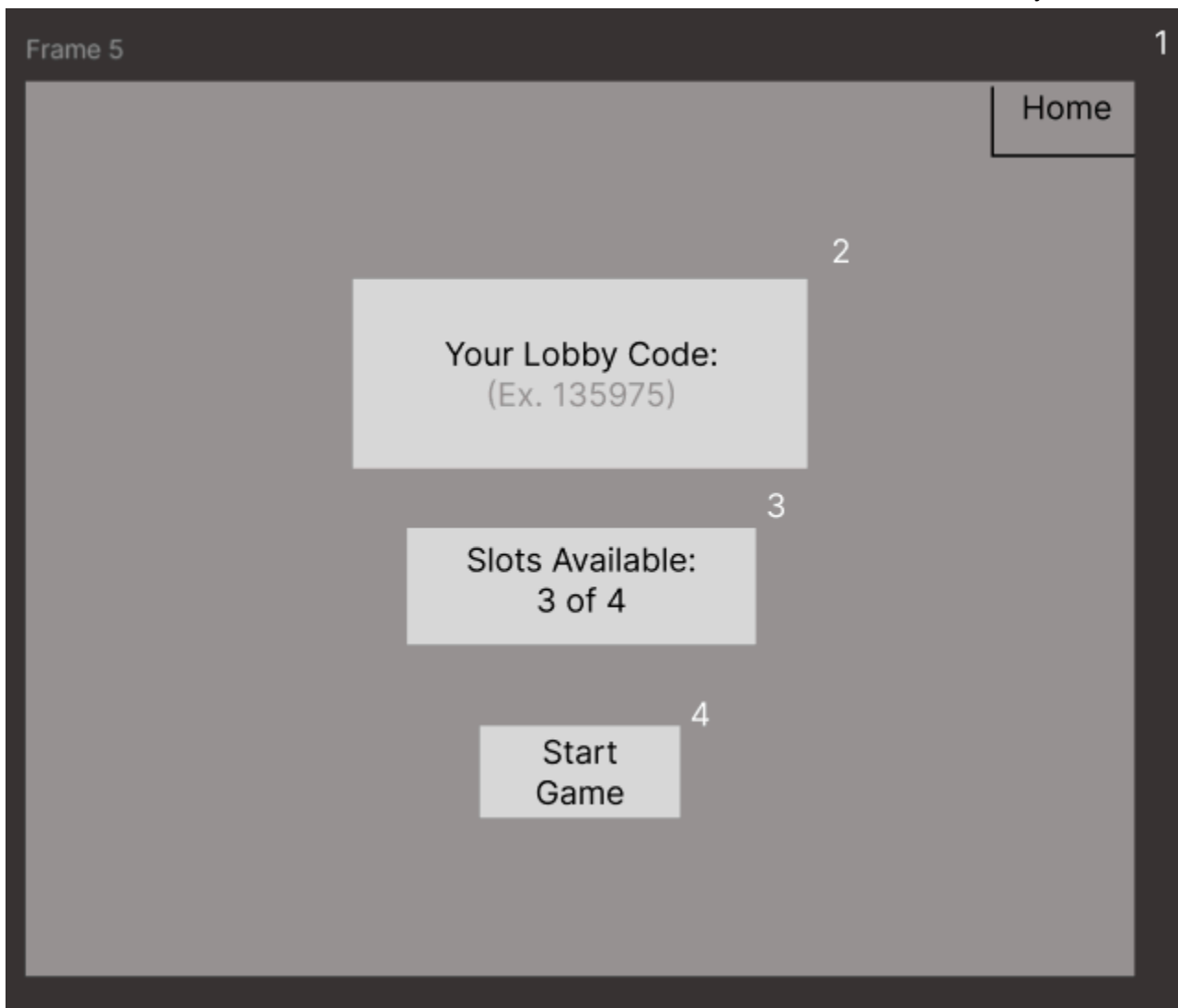


- 1) This is the return home button, should the user want to not play after all
- 2) When multiple players want to play a game someone will need to create a lobby
- 3) When multiple players want to play a game others will need to join a lobby
- 4) When there is one player and no lobby is needed

After login is successful the players will need to decide if they want to play alone or with friends. If with friends we have a way to create and join game lobbies. If alone that is an option too. (This is a subsequent screen only for the player login type.)

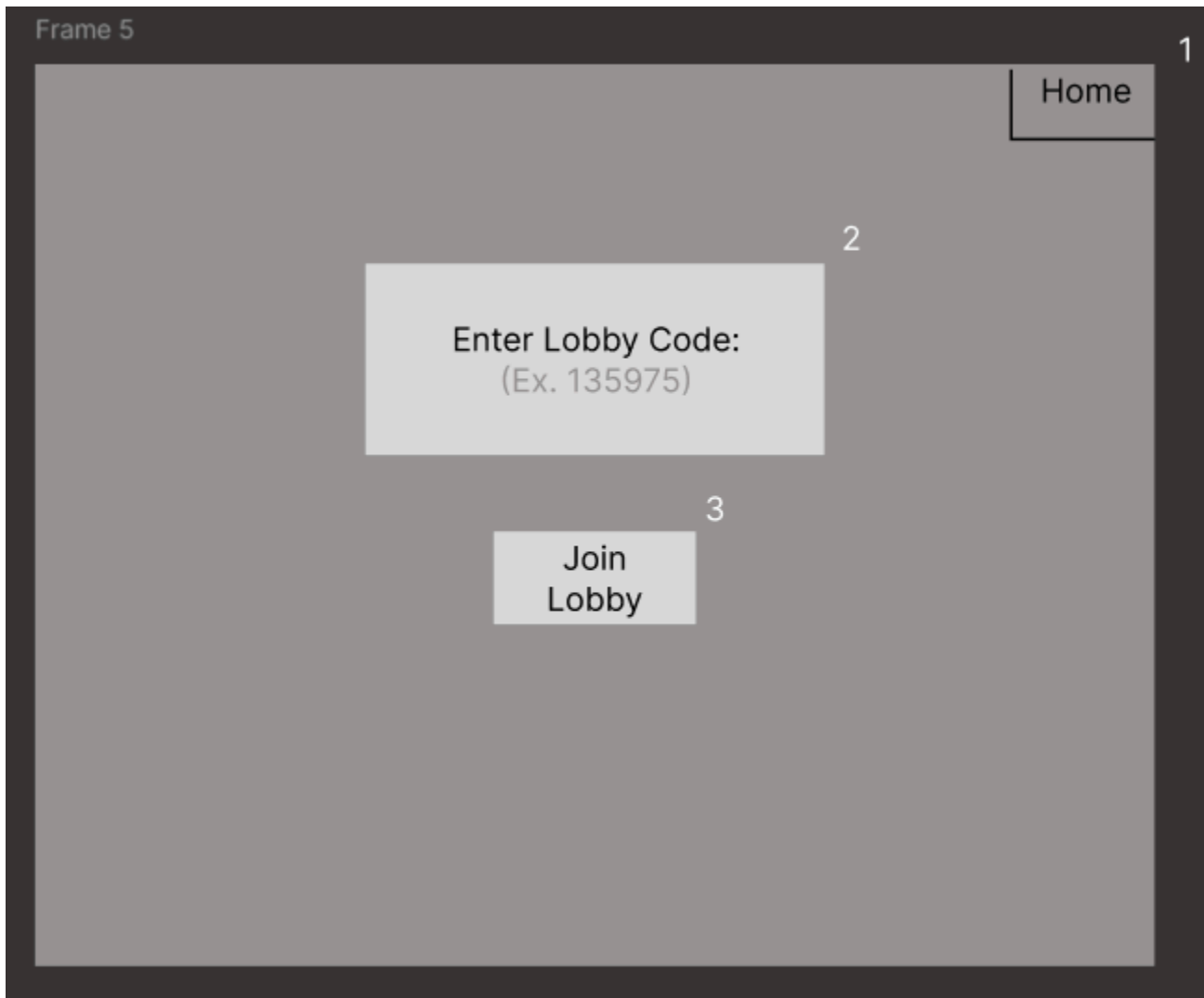


1) This is the generic login success page for all login types
After a successful login there will be a display that welcomes the player. The intent is this will last a certain amount of time on screen then take the user to the next login specific page.



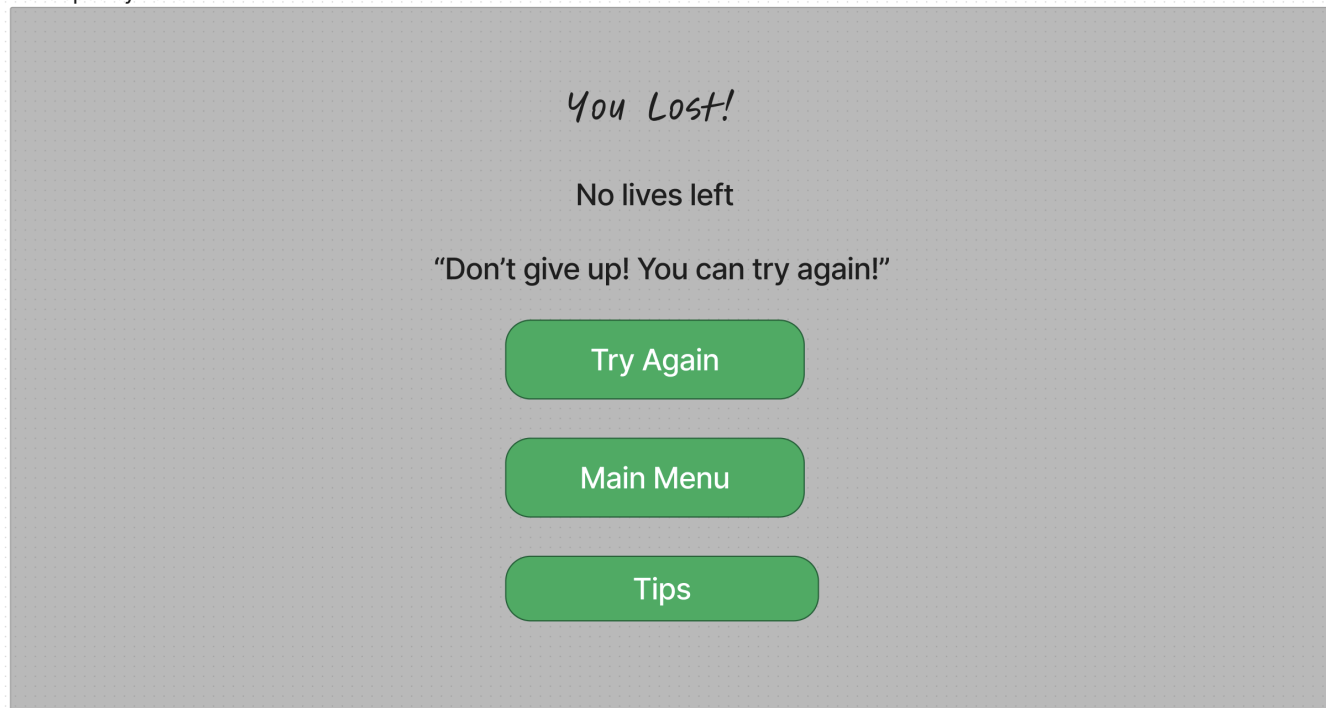
- 1) This is the return home button, should the user want to not play after all
- 2) This displays the randomly generated lobby code that lobby host can then share however
- 3) This displays the number of open lobby slots available, max of 4 players including host
- 4) Once the host has a lobby of whatever size they want they can begin the game

If the player decides to host a game they will create a lobby. This is the page that shows they have created a lobby, the number of vacancies in the lobby, and the code needed for others to join the lobby. Clicking the start game button will take all players in the lobby to the character selection page.



- 1) This is the return home button, should the user want to not play after all
- 2) This is the box where the player can enter the code to join a lobby
- 3) Once the code is entered this button is clicked to try and join the lobby

This page is for players that want multiplayer mode but do not want to create a lobby. They will have to enter the lobby code that the host shares (by whatever means they choose) and then click Join Lobby. If they enter the wrong code we will display some message saying invalid lobby code, else they will get a message saying they joined or tell them the lobby size out of 4 that they joined.



This page opens if the player won the game, here are two buttons which navigates to start the game again or go to main menu or go to tips.

Page 1: You Lost

- The "You Lost" page is crafted to offer a considerate and encouraging experience for players who haven't achieved success in the game.
- Offers motivational messages to uplift players and reduce frustration, reinforcing the idea that improvement is possible.
- The "Play Again" button in a game is designed to invite players back into the action with ease. This button allows you to quickly retry a level or challenge, encouraging you to improve your score or refine your strategy.
- The "Main Menu" allows players to return to the main menu, providing access to various game options. Players can select new game modes, load saved games, adjust settings, view achievements, or check leader boards.
- Tips provide players with helpful advice on game mechanics, strategies, and objectives, making it easier to understand complex elements.

Level Complete!

SCORE : 10000

Bonus : x1.5

Total : 15000

Play Again

Next Level

Main Menu

Page 2: Level Complete

- In this game, reaching the "Level Complete" screen is a moment of celebration! Once you finish a level, you're greeted with an engaging summary that highlights your performance. The screen prominently displays your score, showcasing the points you earned through game play, as well as any bonus points collected along the way for completing challenges or gathering hidden items.
- The "Play Again" button in a game is designed to invite players back into the action with ease. This button allows you to quickly retry a level or challenge, encouraging you to improve your score or refine your strategy.
- The "Main Menu" allows players to return to the main menu, providing access to various game options. Players can select new game modes, load saved games, adjust settings, view achievements, or check leader boards.

Bonus Points

1. Minimalists

Finish a level using the fewest jumps possible.

Bonus: x1.5

2. Life Saver

Finish a level without losing any life.

Bonus: x2

3. Enemy Avoidance

Complete a level without defeating any enemies

Bonus: x2.5

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Page 3: Bonus Points

- Bonus points are additional points awarded to players for achieving specific in-game milestones or completing challenges. Points gained by completing specific achievements or objectives within the game.
- They encourage players to explore the game more thoroughly and engage with its mechanics, promoting replayability. Bonus points contribute to the overall score, allowing players to compare their performance with others. they provide positive reinforcement, rewarding players for their efforts and enhancing the enjoyment of the game.
- In the page Bonus Points page, the player gets bonus points if he/she completes the tasks mentioned such as life saver, enemy avoidance and minimalist.

How to play?

Home

1. Movement

"Use arrow keys to move left and right."

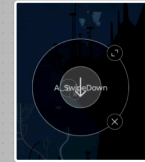
2. Objective

"Avoid obstacles and save lives to win."

3. Tip

"Avoid the attacks by Donkey Kong."

"Use hammers to clear obstacles."



Back

This slide guides us how to play

Page 4: How to play?

- The guide or tutorial section of a game serves as an introductory resource for players, helping them understand the game's mechanics, controls, and objectives.
- Concise explanations of game mechanics, controls, and objectives.
- Tips provide players with helpful advice on game mechanics, strategies, and objectives, making it easier to understand complex elements.